Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/21/2006
PI Name:	Williams, James G Ph.D.		
Project Title:	Improving lunar laser ranging tests of gravitational theory		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	James.G.Williams@jpl.nasa.gov	**Fax:**	FY 818/393-4965
PI Organization Type:	NASA CENTER	**Phone:**	818/354-6466
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	Mail Stop 238-600		
PI Address 2:	4800 Oak Grove Drive		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Turyshev, Slava (Jet Propulsion Laboratory) Boggs, Dale (Jet Propulsion Laboratory)		
Grant/Contract No.:			
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obsacurec by Earth's gravity		

Task Description:	Description: Precise analyses of accurate ranges to the Moon test several aspects of gravitational theory: the Equivalence Principle, geodetic precession, PPN parameters beta and gamma, and possible variation of the gravitational constant G. The uncertainties of these tests decrease as data accuracies improve and data time span lengthens. This project seeks to improve the Lunar Laser Ranging (LLR) models to exploit the University of California at San Diego/University of Washington APOLLO effort's mm accuracies for improved tests of fundamental and gravitational physics. The goals of this project are three-fold: 1) Develop and implement the multidisciplinary modeling and data analysis improvements necessary to advance LLR from the current cm range accuracy to the expected mm accuracy of the APOLLO effort; 2) Support this new ranging effort, including tests for data accuracy and short- and long-term stability; 3) For future space exploration, investigate the performance characteristics which are desirable for the next generation of laser ranging, the fundamental questions which can be investigated, and the challenges for modeling and data analysis. Objective: Precise ranges to the Moon must be accurately analyzed in order to extract the scientific information on the nature of gravity. Accurate models for the computation of the orbit of the Moon and the rotation and orientation of the Earth and Moon, and their realization in computer programs, are needed to do this. This task improves the models and computer programs used for analyzing the Lunar Laser Ranging data. The intent is to improve the accuracy of the data analysis by an order of magnitude in order to match the mm accuracy of the data expected from the new APOLLO facility. The analysis of the highly accurate lunar range data would be used to study the nature of gravitation. Significance: Einstein's theory of gravity has been tested by Lunar Laser Ranging and other techniques. This task would improve the accuracy of LLR tests by an order of magnitude. Physics is seeking the next theory of gravity beyond Einstein's theory. The accuracy of the range data is being improved to provide more accurate data on the nature of gravity. An improvement in data accuracy requires an improvement in analysis accuracy if this exploration of gravitation is to be realized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fundamental research on gravitational physics aids the search for new physics. Ephemeris of Moon is publicly available and is used by space missions. Data is also sensitive to geophysical/geodetic parameters.
Task Progress:	A description of progress with greater detail can be found in the annual report. In summary: a) A chapter in the Explanatory Supplement to the Astronautical Almanac is published. A paper on LLR tests of the equivalence principle remains in press and two papers with J. Mueller have been submitted and are in press. A draft has been prepared of a new paper on ranging to the Moon and beyond. Two presentations were given by S. G. Turyshev and one by J. G. Williams. See the bibliography for new paper and abstract references. b) The first Apache Point ranges were fit. Ranges are in October, November and December 2005 and January 2006. Residuals were sent to Tom Murphy (UCSD). c) A change of relativistic scale for Earth and Moon sites was made in our range processing program. The constant part of the scale change can be viewed as a change of definition, but there is also a 1 mm annual effect. d) Code was modified to add new stations. Station coordinates and coordinate rates can be fit. Apache Point and Matera are the new LLR stations added. e) A shift of computers is necessary and supporting modifications have been started.
Bibliography Type:	Description: (Last Updated: 10/31/2007)
Abstracts for Journals and Proceedings	Mueller J, Williams JG, Turyshev SG. "Lunar Laser Ranging Contributions to Relativity and Geodesy." 359th WE-Heraeus Seminar on “Lasers, Clocks, and Drag-Free: Technologies for Future Space Exploration in Space and Tests of Gravity,“ ZARM, Bremen, Germany, May 30-June 1, 2005. General Relativity and Gravitation, 2006 in press , Dec-2006
Abstracts for Journals and Proceedings	Williams JG. "Lunar Laser Ranging and the Evolution of Lunar Dynamics." 37th AAS Division on Dynamical Astronomy meeting, Halifax, Nova Scotia, Canada, June 25-29, 2006. Bull. Amer. Astron. Soc., in press, July 2006. , Jul-2006
Books/Book Chapters	Williams JG, Turyshev SG, Boggs DH. "Lunar Laser Ranging Tests of the Equivalence Principle with the Earth and Moon." in "Testing the Equivalence Principle on Ground and in Space." Ed. C. Laemmerzahl, C. W. F. Everitt, and R. Ruffini. Springer-Verlag, in press 2006., Jul-2006
Books/Book Chapters	Standish EM, Williams JG. "Orbital Ephemerides of the Sun, Moon, and Planets." in "Explanatory Supplement to the Astronautical Almanac." Ed. Seidelmann, P. K. Washington, DC : U.S. Naval Observatory, 2006. ISBN 1-891389-45-9., Jan-2006
Papers from Meeting Proceedings	Mueller J, Williams JG, Turyshev SG, Shelus PJ. "Potential Capabilities of Lunar Laser Ranging for Geodesy and Relativity." Dynamic Planet 2005: Monitoring and Understanding a Dynamic Planet with Geodetic and Oceanographic Tools, Joint Assembly of International Associations: IAG, IAPSO and IABO, Cairns, Australia, 22-26 August 2005. Joint Assembly of International Associations: IAG, IAPSO and IABO proceedings, August 2005. , Aug-2005